ML12 Show your first chart and choose your project question

Week 4, Clean, chart, question | Session 12 of 24 | Machine Learning

Today's question. Can I take a table from a file to a labelled chart and a careful finding on my own?

Objective. By the end: you can load a table, check it, total it, chart it, write two findings and a limitation, and ask one prediction question a table could answer.

> **WHERE THIS IS USED**
> Every data job starts with one clear question and a table. A coffee trader, a football analyst and a clinic manager each begin that way. Which question would you like your own project to answer?

Before this session: ML11 Charts that show what the numbers say. If you cannot explain the last example, use its error card with the instructor first.

Words for today

Word	What it means	Think of it like this
show-and-tell	A short, friendly turn to show the group something you built.	Showing a friend your drawing.
reproducible	Someone can rerun your notebook from the top and get the same results.	A recipe anyone can follow.
data dictionary	A list of each column with its meaning and unit.	The key at the bottom of a map.
target	The value you want to predict, such as monthly rent.	The number you are guessing.
input	A measurement known before you predict the target.	The clues you are given.
limitation	What the data or method cannot tell you.	The edge of the map.

Block A. The first-chart skeleton

Run the skeleton, then repeat each step in your own notebook, ML12.ipynb, with your own wording.

Predict. Predict the row count, the total revenue and the best-selling item by revenue. Write your guess on paper first.

Your guess, written before you run it

```
import pandas as pd
import matplotlib.pyplot as plt
shop = pd.read_csv("camp_shop.csv")
missing = shop.isna().sum().sum()
shop["revenue_ugx"] = shop["units"] * shop["unit_price_ugx"]
by_item = shop.groupby("item")["revenue_ugx"].sum()
```

```
print(shop.shape[0], missing, by_item.sum())
assert by_item.sum() == shop["revenue_ugx"].sum()
plt.bar(by_item.index, by_item.values)
plt.ylabel("Revenue (UGX)")
plt.title("Revenue by item, practice shop")
plt.show()
```

Run it, then compare. The skeleton prints 60 rows, 0 missing cells and a total revenue of 550000. The assert passes silently because the item totals add up to the overall total. The chart shows Book as the tallest bar. Your own notebook needs the same steps with your two findings, one limitation, a missing-value check and a clean rerun from a restarted kernel.

Investigate.

1. What does the passing assert prove?
2. Why record that there were no missing values?

Modify. Add a finding with a number and one limitation as text cells under the chart.

Block B. Write your project question

Start from the topic on your progress map. In the last 20 minutes choose the supplied table that fits it best and write three sentences: the question, the dataset and the target. Model building starts next week.

Predict. Before you look at the datasets again, write the question you would most like a model to answer. Write your guess on paper first.

Your guess, written before you run it

Run it, then compare. Your three sentences name a question a table can answer, one supplied dataset (synthetic_homes.csv, synthetic_fares.csv or the iris flowers), and the target to predict. Inputs must be known before the target is. Good: predict monthly rent from area and rooms. Weak: explain why rents are high.

Investigate.

1. Which column is the target in synthetic_fares.csv?
2. Why does a prediction question need inputs known beforehand?

Modify. Rewrite a question that asks why into one that asks for a prediction.

Make

In ML12.ipynb report the row count, the revenue by item, one chart, two findings and one limitation, and record your missing-value check. Run all cells from a restarted kernel and save. Add your three-sentence project question. Start from the topic on your progress map and pick the supplied table that fits it best. If none fits, say so; real public data is used only after you and the instructor have read its licence.

Save your work as ML12. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

Error card

You see	It means	Fix
The notebook works only until I restart it	A later cell uses a name that an earlier cell created and you deleted.	Restart the kernel and run all cells from the top to find the missing step.
The totals do not agree	A filter or a typo dropped rows before grouping.	Compare the row counts before and after each step.

Exit ticket

My first chart takes a table from a file to a labelled picture and a careful finding that anyone can rerun.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or build tasks. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been shared until you use the route in START_HERE.pdf.

After this session: foundation lab F04 on the fourth day of the week.